

## Deep Learning-Based Wireless Signal Modulation Classifier Using Synthetic IQ Data

This project focuses on the development of an intelligent wireless signal modulation classification system using deep learning techniques and synthetic In-phase and Quadrature (IQ) data. Automatic Modulation Classification (AMC) is a critical task in modern wireless communication systems, cognitive radio networks, spectrum monitoring, electronic warfare, and next-generation 5G/6G technologies. The objective of this project is to accurately identify the modulation scheme of a received wireless signal without prior information about the transmitter.

The project begins with the generation of synthetic IQ data representing different digital modulation schemes. Synthetic data was used because it allows controlled experimentation under varying signal conditions and helps create a balanced dataset for model training. Various modulation types can be simulated with different signal-to-noise ratio (SNR) levels to improve the robustness of the classification model.

After data generation, the IQ samples are preprocessed and formatted for deep learning. The dataset is then used to train a neural network capable of learning complex patterns embedded within wireless signals. Unlike traditional signal processing approaches that depend heavily on handcrafted features, deep learning automatically extracts relevant features from raw IQ samples, improving classification performance and reducing manual effort.

The model architecture is designed to recognize unique characteristics of modulation schemes and classify incoming signals with high accuracy. During training, the network learns to distinguish subtle differences between signal patterns, even in noisy environments. Performance is evaluated using standard machine learning metrics such as accuracy, loss curves, and confusion matrices to analyze the model's effectiveness.

One of the major advantages of this project is its relevance to future wireless communication systems. With the rapid growth of connected devices and increasing spectrum congestion, intelligent spectrum awareness has become essential. Automated modulation classification can support adaptive communication systems, dynamic spectrum access, interference detection, and secure wireless operations.

The project demonstrates the practical application of Artificial Intelligence and Deep Learning in the telecommunications domain. It combines concepts from wireless communications, digital signal processing, machine learning, and neural networks, providing an interdisciplinary solution to a real-world problem. The implementation showcases how data-driven approaches can outperform conventional techniques in

complex signal environments.

Future enhancements may include training on real-world radio frequency datasets, experimenting with advanced architectures such as CNNs, LSTMs, and Transformers, and deploying the model for real-time modulation recognition. The project serves as a strong foundation for research in AI-enabled wireless networks, cognitive radio systems, and intelligent communication technologies.

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